

Assignment 1 Reflection

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Briefly, I have a tendency to run long, so I apologize in advance. I have tried to keep this to two pages but I have lots of excess thoughts here about stuff so I am sorry it went long.

How I designed the agents and why

I started with the class material and the supplied hints. I have much more experience building software than writing university-level research reviews, so I used your provided structures as a starting point and worked through what each agent needed to contribute. I separated query expansion, source discovery, synthesis, and writing, as directed. The Query Expander develops sub-questions and search terms; the Source Hunter retrieves evidence; the Synthesizer looks for patterns, agreements, contradictions, and gaps; and the Report Writer turns that analysis into a literature review. In every one of those cases, the definitions were written in line with your suggestions.

The goals and backstories reinforce those differences in an attempt to ensure the process is comprehensive. The investigator is told to vary searches and preserve attribution, while the analyst is told to compare evidence without forcing consensus. Only the Source Hunter has the `search_papers` tool. The tasks specify expected outputs and explicit context dependencies: retrieval receives the search strategy, synthesis receives the strategy and evidence, and writing receives all three earlier outputs. A sequential crew makes those handoffs straightforward. The tradeoff is that a weak search constrains everything after it. A single agent would reduce handoffs, but would make those responsibilities harder to inspect; a review-and-retry loop would offer more control at greater cost.

I will be candid and say I understand the output (though I can see from the PDFs that the Report Writer needs some guidance about using special characters, since they are all through the documents), and I understand how the agents did their work (to a large extent, though parsing all those screens took some doing for me to make sense of), and I get the crew concept, I am not sure my agent definitions were as optimized as they could be. I am not sure what's happening at the mechanical layer but I couldn't help but feel like maybe the Source Hunter shouldn't be using sub-agents with different context for each of the subquestions brokered out by the query expanded, and then delivering those results in a concurrent fashion instead of sequential. I know the question about production implementation is later, but increasing the size of the RAG to production levels and keeping this particular crew of agents to do research by themselves feels like it would probably take a long time to return, which may be fine, but feels problematic.

The pipeline produced useful reports, so I think from that perspective the agents were, if not well-designed, at least functionally designed. It also showed me where a research workflow needs more control than a good set of prompts.

It also generated very research-specific reports, which is great given the assignment, but I suspect that in practice, I probably would have preferred a larger executive summary or a key concepts sub-section or something and then state the other details later as it currently does in the findings section, though that may be a standard presentation methodology that was used.

Consistently the system called out failures in the source data; in fact, to the point, that I re-ran the verify several times to make sure I hadn't inadvertently corrupted something. I would imagine that an agent with a web crawler tool of some kind would have been useful particularly

given an instruction to clean up or complete an excerpt by finding the original paper, and in the context of this crew, if you used it solely in that way, and identified in the resultant paper what you had done, I think that would probably have had substantial additional value.

One thing I specifically added to the report writer was the “You always write all output using ASD-STE100 standards.” instruction, because I find it keeps the excess down. I did use GPT 6.0 Luna instead of the 4o-mini in the example just because I am more used to it and I think it’s fast enough and hallucinates less (though, thinking about it, maybe you wanted hallucinations; I might run it again with one of the advanced questions with that model instead); plus, the 58K long item exceeded my immediate token limit:

```
'type': 'tokens', 'param': None, 'code': 'rate_limit_exceeded'}}}.
Retrying in 1s...
```

```
OpenAI API error: Error code: 429 - {'error': {'message': 'Request
too large for text-embedding-3-small in organization
```

```
org-th5w4k5pCRFTs9S6XslcqqM5 on tokens per min (TPM): Limit 40000,
Requested 58098. The input or output tokens must be reduced in order
to run successfully.
```

And that drove me down a path of changing the python settings, etc., but that slowed the processing WAY down, so I went back to my API configurations on the Open AI side and realized I had to tell it to support a larger message per minute count, and then it ran fine.

ReAct-style reasoning in the outputs

As per the instructional materials, ReAct connects reasoning, an action, and an observation that informs the next decision. In this system, the clearest place for that loop is source discovery: identify an evidence need, call `search_papers`, inspect the returned passages, and decide whether another query is needed. The task explicitly asks for varied searches, but the three saved reports do not preserve individual tool calls or the sequence of decisions. I cannot honestly present them as a complete ReAct execution trace; however, you can see the ReAct loop if you troll back through the output because it has task start / task completion bodies and within that (for instance, in the `search_research_papers` tool used by the Source Hunter agent, you can see in those captions where the tool execution is counted, and and between each loop it logs the arguments called, and the arguments proceed down the cycle from the query broken out by the Query Extender. In the case of the “How Do Multi-Agent Systems Coordinate and Communicate?” question, the Search Hunter executed the tool it had available 15 times, applying observation before action (the specific of the query followed by the tool call) .

The output, aside from the log grepping I did by hand, shows evidence-aware decisions consistent with that workflow. In “How Do Multi-Agent Systems Coordinate and Communicate?”, Findings 4 observes that both systems in the retrieved AutoGen comparison answered eight of ten questions correctly, then declines to claim a general multi-agent accuracy advantage. Findings 5 narrows the answer because the retrieved evidence does not adequately cover classical coordination mechanisms. In “What Is an AI Agent?”, the Discussion identifies fragmentary and garbled excerpts, and the Conclusion treats its definition as a working description rather than a universal standard. These are specific examples of conclusions responding to evidence limits, although they do not prove an iterative tool-use loop. The approaches report explains ReAct in Findings 3; explaining the method is also not evidence that my agents executed it. I would save the query, tool-result, and follow-up decision records that were generated by the run into a verifiable ledger to make that distinction verifiable.

What worked, what did not, and what I would change

The role separation and context wiring gave the reports a coherent structure. All three include the required sections, and the coordination report does more than summarize: it explains why simulation measures and question-answer accuracy cannot be compared directly. The system also admits missing evidence. That part worked better than I initially expected.

The retrieval interface is a more concrete weakness than simply having too small a corpus. It truncates each passage to 500 characters and labels a rough conversion of vector distance as relevance. The reports repeatedly acknowledge incomplete excerpts, which is consistent with this limitation. The displayed metadata does not provide precise passage-level page locations or complete bibliographic records. A citation callout is useful, but it does not establish that the cited source supports the claim. ***There is also a visible citation mismatch in the approaches report: the autonomous-agent survey is attributed to Xi et al. in the body and Wang et al. in the references.***

There are a number of things I would change here:

- Constrain query expansion with a sub-question budget, priorities, and coverage checks.
- Define synthesis more explicitly:
 - compare claims, methods, conditions, and results;
 - separate genuine disagreement from different evaluation settings;
 - distinguish missing retrieval evidence from a gap in the literature.
- Return fuller passages with stable source identifiers and add a citation check before writing.

The Markdown structure is largely acceptable, I think, though I would probably change the description of that agent a bit to force it to remove repeated qualifications and strive for a clearer answer.

The writer's ASD-STE100 instruction may contribute to a restricted style, which I generally find cleaner to work with, but in this case, it may be inappropriate given the form that its trying to generate into. However, I do not believe that alone would cause most of these gaps; they feel like more constructive problems rather than prose and alignment problems.

HOWEVER, running it again against the last question while using the 4o-mini model resulted in a markedly worse paper, in my opinion, while running it again using that model with the ASD-STE100 instruction removed generated a more complete paper, I think. I suspect that removing that and using it with Luna instead of Mini would also have given me a more satisfactory result, in my opinion. I don't particularly like the bullet-pointed methodology that mini is using, I much prefer the prose-heavy methodology that Luna generated, I found it made it far easier for me to read the reports.

What production would require

A larger RAG corpus would help, but only with better extraction, retrieval, and provenance. Production would need reliable paper metadata, exact passage locations, validated relevance thresholds, and an explicit insufficient-evidence result. I would add human checkpoints after retrieval and synthesis so a reviewer can inspect the evidence and redirect the work before the final report. I am truly unsure from here whether some of the limitations I have described are a function of the agents as defined, the RAG corpus, or some mismanagement of crew and/or Report Writer instructions on my part.

I would also want tests for retrieval coverage and citation support, evaluations against simpler baselines, recorded task and tool outputs, retry and timeout handling, cost limits, and versioned prompts and corpus data. Memory needs isolation between research runs, and retrieved text

needs to be treated as source material rather than instructions. The central lesson for me is that a readable report is only one part of the result. Before using this system for decisions, I need to be able to verify how it reached its claims and trust it to stop when the evidence is not there.